

MATH 123 LECTURE NOTES

WEEK VIII

INTEGRATION

SUMS AND SIGMA NOTATION

Sigma Notation

→ If m and n are integers with $m \leq n$, and if f is a function defined at the integers $m, m+1, m+2, \dots, n$, the symbol $\sum_{i=m}^n f(i)$ represents the sum of the values of f at those integers:

$$\sum_{i=m}^n f(i) = f(m) + f(m+1) + f(m+2) + \dots + f(n)$$

→ The explicit sum appearing on the right side of this equation is the expansion of the sum represented in sigma notation on the left side.

Ex 7 $\sum_{i=1}^5 i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

→ Sometimes we use a subscripted variable a_i to denote the i th term of a general sum instead of using the functional notation $f(i)$:

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + a_{m+2} + \dots + a_n$$

$$\rightarrow \sum_{i=m}^n (A f(i) + B g(i)) = A \sum_{i=m}^n f(i) + B \sum_{i=m}^n g(i)$$

$$\sum_{j=m}^{m+n} f(j) = \sum_{i=0}^n f(i+m)$$

(1)

$$(*) \quad S = \sum_{i=1}^n i = 1+2+3+\dots+n = \frac{n(n+1)}{2}$$

$$S = 1+2+3+\dots+(n-1)+n$$

$$S = n+(n-1)+(n-2)+\dots+2+1$$

$$2S = \underbrace{(n+1) + (n+1) + (n+1) + \dots + (n+1)}_{n \text{ times}} = n(n+1)$$

$$S = \frac{n(n+1)}{2}$$

Three Summation Formulas

$$(a) \quad \sum_{i=1}^n 1 = \underbrace{1+1+\dots+1+1}_{n \text{ times}} = n$$

$$(b) \quad \sum_{i=1}^n i = 1+2+3+4+\dots+(n-1)+n = \frac{n(n+1)}{2}$$

$$(c) \quad \sum_{i=1}^n i^2 = 1^2+2^2+3^2+4^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(d) \quad \sum_{i=1}^n r^{i-1} = 1+r+r^2+\dots+r^{n-1} = \frac{r^n-1}{r-1} \text{ if } r \neq 1$$

Proof of (c) To prove (c) we write n copies of the identity $(k+1)^3 - k^3 = 3k^2 + 3k + 1$

one for each value of k from 1 to n , and add them up

$$2^3 - 1^3 = 3 \cdot 1^2 + 3 \cdot 1 + 1 \quad 3 \cdot \sum_{i=1}^n i^2 = 3n^2 + 3n + 1 - n - \frac{3n(n+1)}{2}$$

$$3^3 - 2^3 = 3 \cdot 2^2 + 3 \cdot 2 + 1$$

$$4^3 - 3^3 = 3 \cdot 3^2 + 3 \cdot 3 + 1$$

$$\vdots$$

$$n^3 - (n-1)^3 = 3(n-1)^2 + 3(n-1) + 1$$

$$(n+1)^3 - n^3 = 3n^2 + 3n + 1$$

$$\frac{(n+1)^3 - 1^3}{3} = \sum_{i=1}^n i^2 + 3 \sum_{i=1}^n i$$

$$= 3n^2 + 3n + 1$$

$$+ n = 3 \sum_{i=1}^n i^2 + \frac{3n(n+1)}{2} + n$$

Ex 7 Evaluate $\sum_{k=m+1}^n (6k^2 - 4k + 3)$ where $1 \leq m < n$.

$$\begin{aligned} \sum_{k=1}^n (6k^2 - 4k + 3) &= 6 \sum_{k=1}^n k^2 - 4 \sum_{k=1}^n k + 3 \sum_{k=1}^n 1 \\ &= 6 \frac{n(n+1)(2n+1)}{6} - 4 \frac{n(n+1)}{2} + 3n \\ &= 2n^3 + n^2 + 2n \end{aligned}$$

$$\begin{aligned} \sum_{k=m+1}^n (6k^2 - 4k + 3) &= \sum_{k=1}^n (6k^2 - 4k + 3) - \sum_{k=1}^m (6k^2 - 4k + 3) \\ &= 2n^3 + n^2 + 2n - 2m^3 - m^2 - 2m \end{aligned}$$

Areas as Limits of Sums

(i) The area of a plane region is nonnegative real number of square units.

(ii) The area of a rectangle with width w and height h is $A = wh$



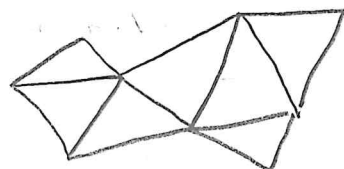
(iii) The areas of congruent plane regions are equal if evaluated.

(iv) If region S is contained in region R , then the area of S is less than or equal to that of R



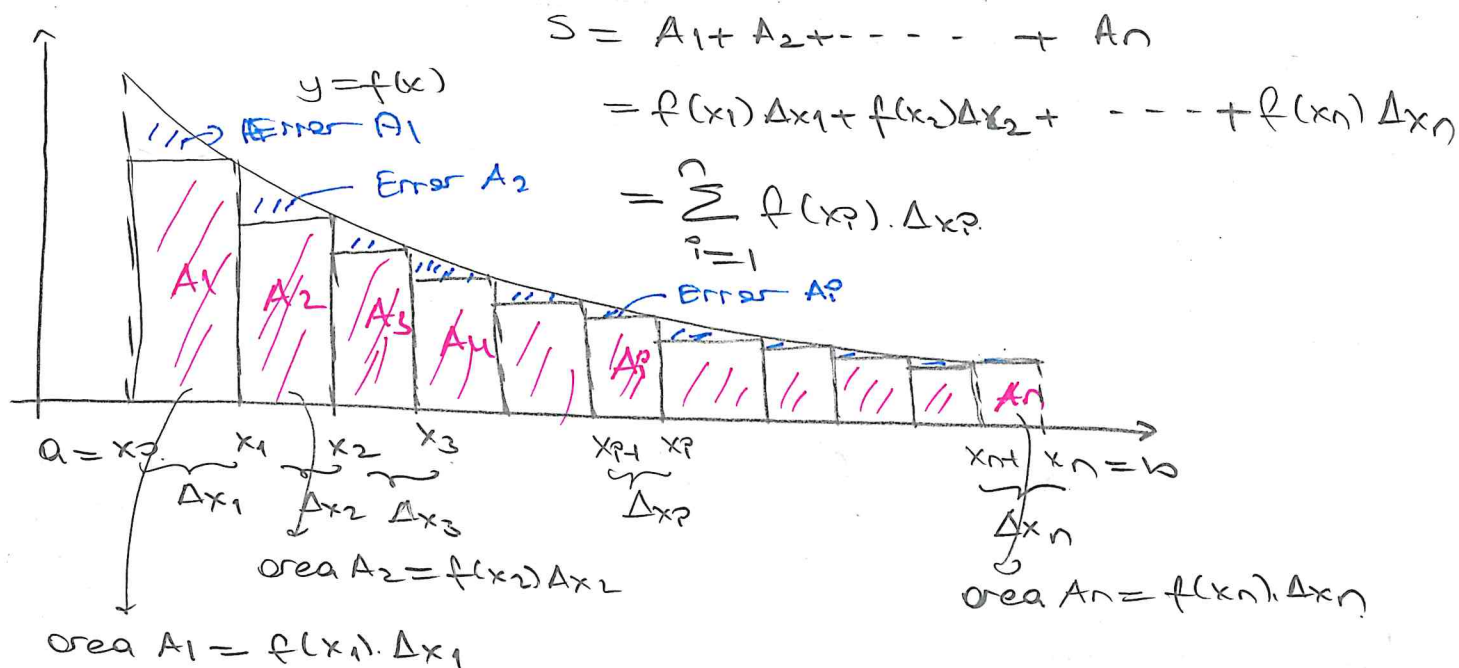
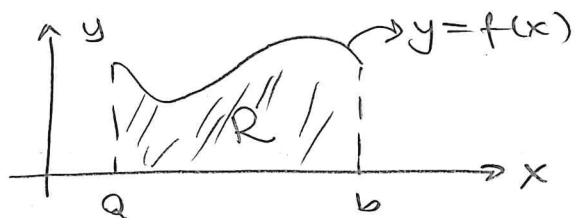
area $S \leq$ area R

(v) If region R is a union of (finitely many) nonoverlapping regions, then the area of R is the sum of areas of those regions



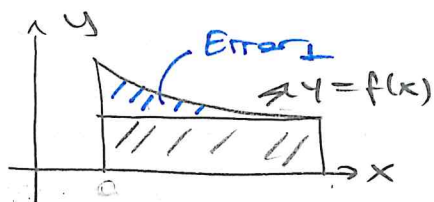
area of polygon =
sum of areas of
trapezoids.

10 The basic area problem: find the area of region R

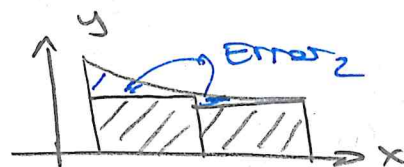


$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_{n-1} < x_n = b$$

$$\Delta x_i = x_i - x_{i-1} \quad (i = 1, 2, 3, \dots, n)$$

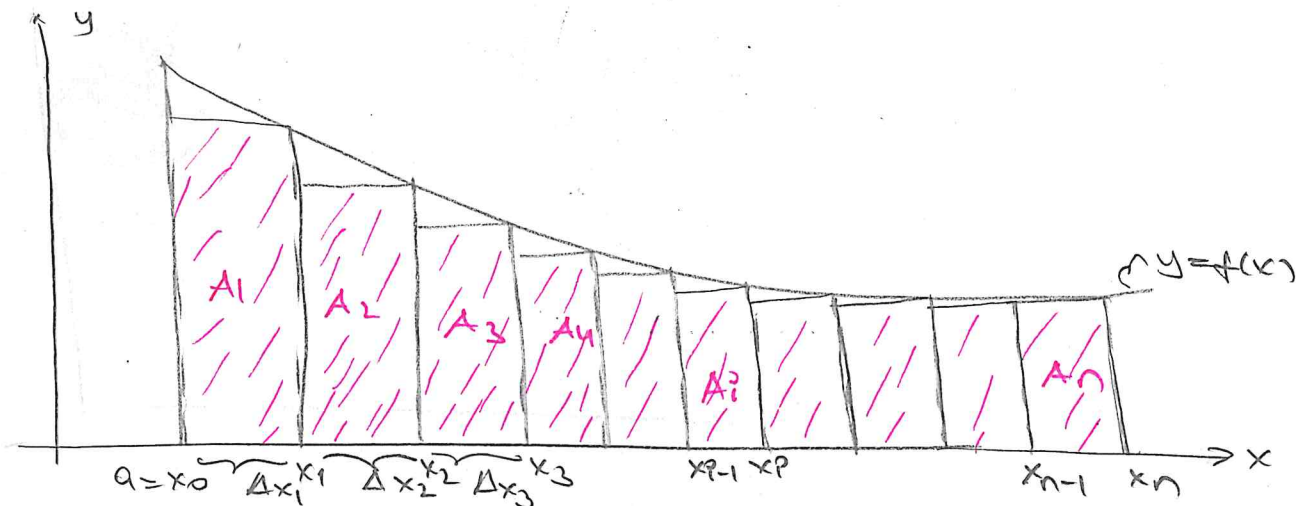


Use more rectangles



$$\text{Error}_2 < \text{Error}_1$$

Using more rectangles makes the error smaller.

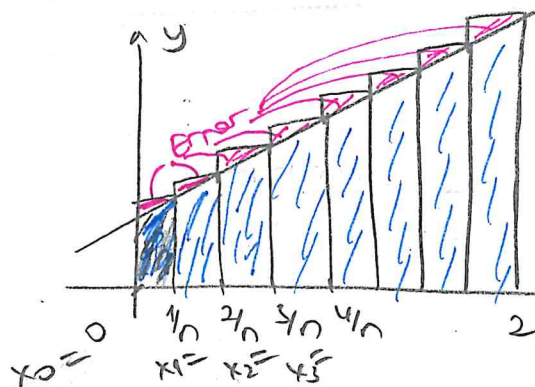


$$S_n = f(x_1)\Delta x_1 + f(x_2)\Delta x_2 + \dots + f(x_n)\Delta x_n$$

$$= \sum_{i=1}^n f(x_i)\Delta x_i$$

Area of $R = \lim_{\substack{n \rightarrow \infty \\ \max \Delta x_i \rightarrow 0}} S_n$

Ex 7 Find the area of the region lying under the straight line $y = x + 1$ above the x -axis between the lines $x = 0$ and $x = 2$.



$$S_n = \sum_{i=1}^n \left(\frac{2i}{n} + 1 \right) \cdot \frac{2}{n}$$

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

$$S_n = \left(\frac{2}{n} \right) \sum_{i=1}^n \left(\frac{2i}{n} + 1 \right)$$

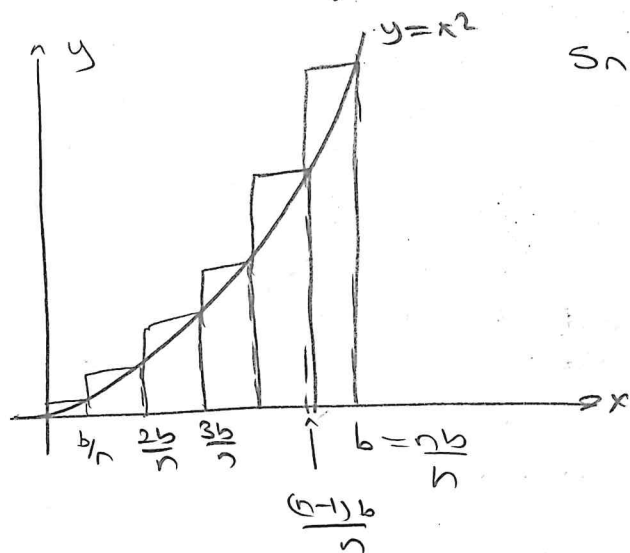
$$= \left(\frac{2}{n} \right) \left[\left(\frac{2}{n} \right) \sum_{i=1}^n i + \sum_{i=1}^n 1 \right]$$

$$= \frac{2}{n} \left[\frac{2}{n} \frac{n(n+1)}{2} + n \right]$$

$$= 2 \left(\frac{n+1}{n} \right) + 2$$

$$A = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(2 \frac{n+1}{n} + 2 \right) = 2 + 2 = 4 \text{ square units}$$

Ex Find the area of the region bounded by the parabola $y=x^2$ and the straight lines $y=0$, $x=0$ and $x=b$ where $b>0$



$$S_n = \sum_{i=1}^n \left(\frac{ib}{n}\right)^2 \cdot \frac{b}{n}$$

$$= \frac{b^3}{n^2} \sum_{i=1}^n i^2$$

$$= \frac{b^3}{n^2} \cdot \frac{(n+1)(2n+1)}{6}$$

hence, the required area is

$$A = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} b^3 \frac{n(n+1)(2n+1)}{6n^2} = \frac{b^3}{3} \text{ square unit}$$

THE DEFINITE INTEGRAL

→ Let P be a finite set of points arranged in order between a and b on the real line, say

$$P = \{x_0, x_1, \dots, x_{n-1}, x_n\}$$

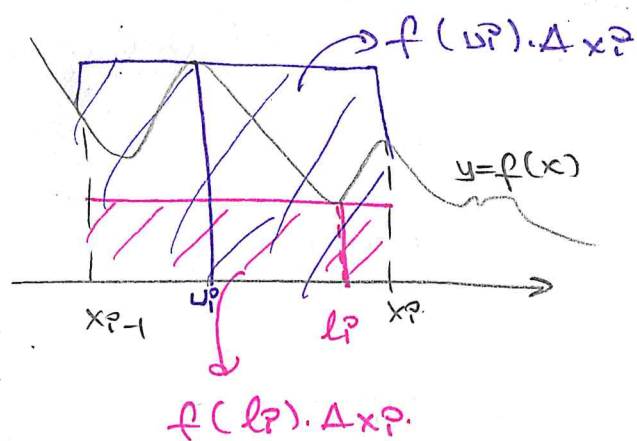
where $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$. Such a set P is called a partition of $[a, b]$. The length of the i th subinterval of P is

$$\Delta x_i = x_i - x_{i-1} \quad (\text{for } 1 \leq i \leq n)$$

and we call the greatest of these numbers Δx_i the norm of the partition P and denote it $\|P\|$:

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i$$

RIEMANN SUMS



→ Since f is continuous on each subinterval $[x_{p-1}, x_p]$ of P , it takes on maximum and minimum values at points of that interval. (Max-Min Theorem)

→ If A_p is that part of the area under $y=f(x)$ and above the x -axis that lies in the vertical strip between $x=x_{p-1}$ and $x=x_p$ then

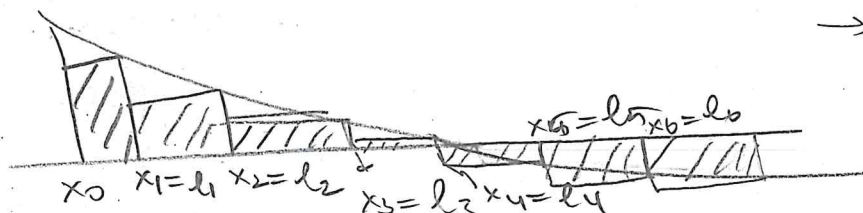
$$f(l_p) \Delta x_p \leq A_p \leq f(u_p) \Delta x_p$$

The lower (Riemann) sum, $L(f, P)$, and the upper (Riemann) sum, $U(f, P)$, for the function f and the partition P are defined by

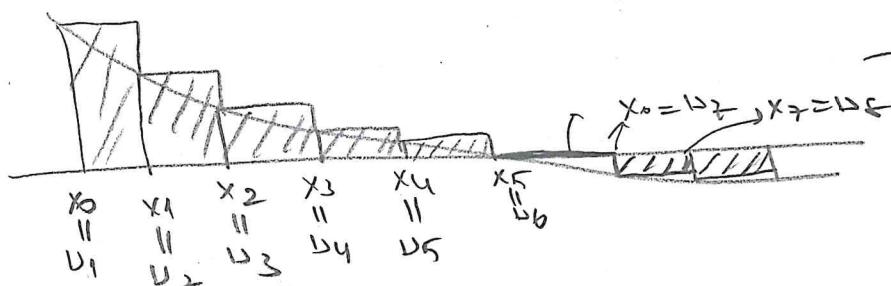
$$L(f, P) = f(l_1) \Delta x_1 + f(l_2) \Delta x_2 + \dots + f(l_n) \Delta x_n = \sum_{i=1}^n f(l_i) \Delta x_i$$

$$U(f, P) = f(u_1) \Delta x_1 + f(u_2) \Delta x_2 + \dots + f(u_n) \Delta x_n = \sum_{i=1}^n f(u_i) \Delta x_i$$

→ A lower Riemann sum and an upper Riemann sum for a decreasing function

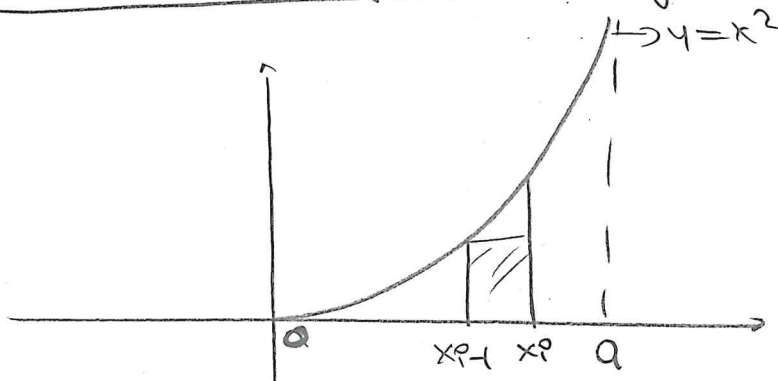


→ $L(f, P)$ for decreasing function.



→ $U(f, P)$ for decreasing function

Ex 7 Calculate the lower and upper Riemann sums for the function $f(x) = x^2$ on the interval $[0, a]$ (where $a > 0$) corresponding to the partition P_n of $[0, a]$ into n subintervals of equal length.



$$\Delta x = \frac{a-0}{n}$$

$$x_i = x_0 + i \cdot \Delta x$$

$$x_0 = 0$$

$$x_i = a \cdot \frac{i}{n}$$

Increasing function
therefore $f(x_i) > f(x_{i-1})$

For each interval

$$x_{i-1} = l_i,$$

$$x_i = u_i.$$

Increasing function. $f(x) = x^2$

$$L(f, P) = \sum_{i=1}^n (x_{i-1})^2 \cdot \Delta x$$

$$= \sum_{i=1}^n \left(\frac{a(i-1)}{n} \right)^2 \cdot \frac{a}{n}$$

$$= \frac{a^3}{n^3} \cdot \sum_{i=1}^n (i-1)^2 = \frac{a^3}{n^3} \cdot \sum_{j=0}^{n-1} j^2 = \frac{a^3}{n^3} \cdot \frac{(n-1)n(2(n-1)+1)}{6}$$

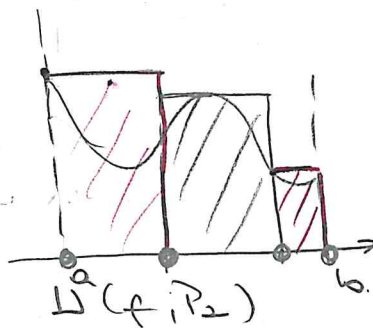
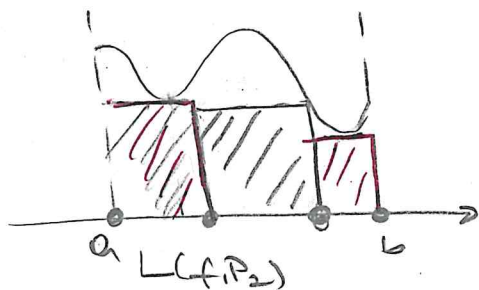
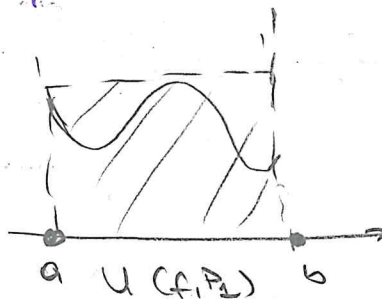
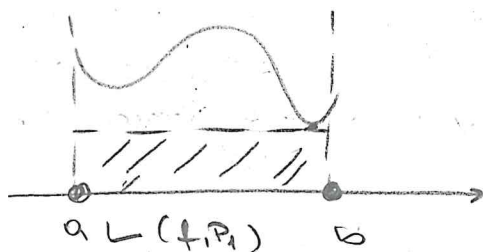
$$= \frac{(n-1)(2n-1)a^3}{6n^2}$$

$$U(f, P) = \sum_{i=1}^n (x_i)^2 \cdot \Delta x$$

$$= \sum_{i=1}^n \left(\frac{a i}{n} \right)^2 \cdot \frac{a}{n} = \frac{a^3}{n^3} \sum_{i=1}^n i^2 = \frac{a^3}{n^3} \sum_{j=0}^{n-1} j^2$$

$$= \frac{a^3}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)a^3}{6n^2}$$

THE DEFINITE INTEGRAL



If P_1 and P_2 are two partitions of $[a, b]$ such that every point of P_1 also belongs to P_2 , then we say P_2 is a refinement of P_1 .



P_2 is a refinement of P_1 .

→ If P_2 is a refinement of P_1 then we have

$$L(f, P_1) \leq L(f, P_2) \leq U(f, P_2) \leq U(f, P_1)$$

→ Also, given any two partitions P_1 and P_2 we can have a common refinement P by combining these partitions. So,

$$L(f, P_1) \leq L(f, P) \leq U(f, P) \leq U(f, P_2)$$

→ Each lower sum is $\leq$ each upper sum

→ Therefore, there is at least one number I such that

$$L(f, P) \leq I \leq U(f, P)$$

for any partition of P

(By completeness of reals).

Definition

The Definite Integral

→ Suppose there is exactly one number I such that for every partition of $[a, b]$ we have

$$L(f, P) \leq I \leq U(f, P)$$

→ Then, we say that the function f is integrable on $[a, b]$ and we call I the definite integral of f on $[a, b]$

The definite integral is denoted by the symbol:

$$I = \int_a^b f(x) dx$$

→ The definite integral of $f(x)$ over $[a, b]$ is a number, it is not a function of x .

Integral sign

$$\int_a^b f(x) dx = \int_a^b f(t) dt$$

Annotations:
- b : upper limit of integration
- a : lower limit of integration
- $f(x) dx$: integrand
- dx : variable of integration

Ex Show that $f(x) = x^2$ is integrable over $[0, a]$ where $a > 0$ and evaluate $\int_0^a x^2 dx$ (see Ex in paper)

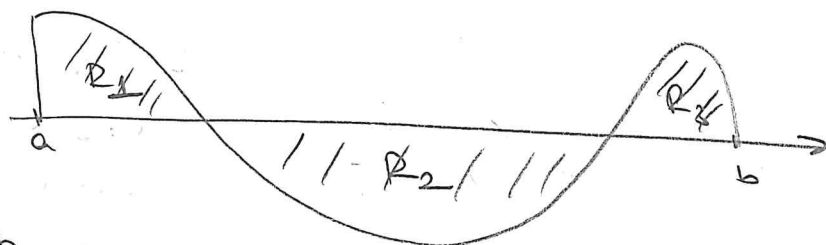
$$\lim_{n \rightarrow \infty} L(f, P_n) = \lim_{n \rightarrow \infty} \frac{(n-1)(2n-1)a^3}{6n^2} = \frac{a^3}{3}$$

$$\lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)a^3}{6n^2} = \frac{a^3}{3}$$

If $L(f, P_n) \leq I \leq U(f, P_n)$, we must have $I = a^3/3$. Thus, $f(x) = x^2$ is integrable over $[0, a]$ and

$$\int_0^a f(x) dx = \int_0^a x^2 dx = \frac{a^3}{3}$$

Note



$$\int_a^b f(x) dx = \text{Area } R_1 - \text{Area } R_2 + \text{Area } R_3.$$

General Riemann Sum

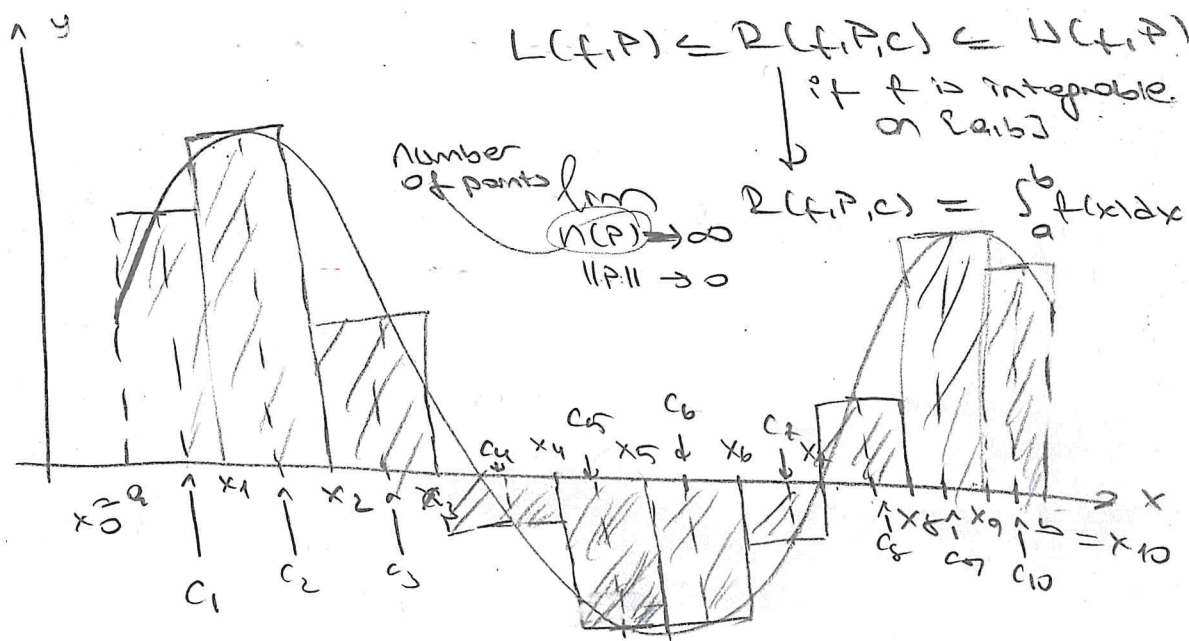
Let $P = \{x_0, x_1, x_2, \dots, x_n\}$, where $a = x_0 < x_1 < x_2 < \dots < x_n = b$ be a partition of $[a, b]$ having norm $\|P\| = \max_{1 \leq i \leq n} \Delta x_i$.

In each subinterval $[x_{i-1}, x_i]$ of P pick a point c_i (called a tag). Let $C = (c_1, c_2, \dots, c_n)$ denote the set of these tags.

The sum

$$R(f, P, C) = \sum_{i=1}^n f(c_i) \Delta x_i = f(c_1) \Delta x_1 + f(c_2) \Delta x_2 + \dots + f(c_n) \Delta x_n$$

is called the Riemann sum of f on $[a, b]$ corresponding to partition P of tags C .



Thm If f is continuous on $[a, b]$, then f is integrable on $[a, b]$

Ex Express the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left(1 + \frac{2^{i-1}}{n}\right)^{1/3}$ as a definite integral.

$$\Delta x = \frac{b-a}{n}$$

$$x_i = a + i \Delta x$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta x f(c_i)$$

$$a = x_0 = 0$$

$$x_1 = \frac{2}{n}$$

$$x_2 = \frac{4}{n}$$

$$x_3 = \frac{6}{n}$$

$$b = x_n = \frac{2 \cdot n}{n} = 2$$

$$x_1 = \frac{2 \cdot 1}{n}$$

$$x_2 = \frac{2 \cdot 2}{n}$$

$$x_3 = \frac{2 \cdot 3}{n}$$

$$x_i = 0 + i \Delta x$$

$$x_i = \frac{2i}{n}$$

your interval

$$[x_{i-1}, x_i] = \left[\frac{2(i-1)}{n}, \frac{2i}{n} \right]$$

$$f(x_i) = \left(1 + \frac{2^{i-1}}{n}\right)^{1/3}$$

$$f(x) = (1+x)^{1/3}$$

observe that

$$c_i \in \left[\frac{2(i-1)}{n}, \frac{2i}{n} \right]$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left(1 + \frac{2^{i-1}}{n}\right)^{1/3} = \int_0^2 (1+x)^{1/3} dx$$

PROPERTIES OF DEFINITE INTEGRALS

Let f and g be integrable on an interval containing the points a, b and c . Then

$$(a) \int_a^a f(x) dx = 0$$

$$(b) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$(c) \int_a^b (A f(x) + B g(x)) dx = A \int_a^b f(x) dx + B \int_a^b g(x) dx$$

(d) An integral depends additively on the interval of integration

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$



(e) If $a \leq b$ and $f(x) \leq g(x)$

for $a \leq x \leq b$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$

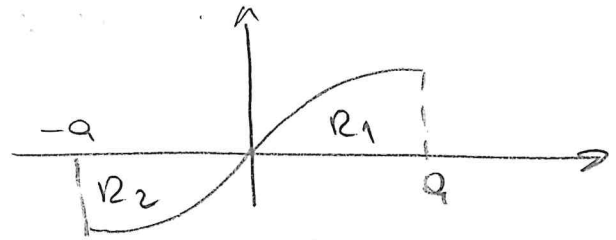
(b)

① Triangle inequality If $a \leq b$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

② If f is an odd function $f(-x) = -f(x)$, then

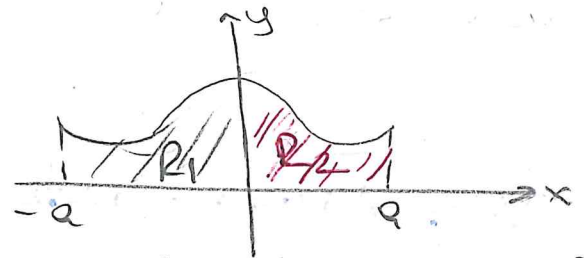
$$\int_{-a}^a f(x) dx = 0$$



$$2 \text{area } R_1 - 2 \text{area } R_2 = 0$$

③ If f is an even function $f(-x) = f(x)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$



$$2 \text{area } R_1 + 2 \text{area } R_2 = 2 \text{area } R_1$$

Ex Evaluate

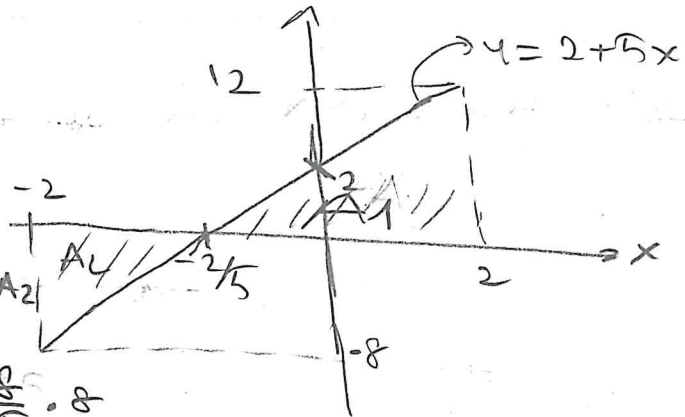
④ $\int_{-2}^2 (2+5x) dx = ?$

I. WAY 2

$$\int_{-2}^2 (2+5x) dx = \text{area } A_1 - \text{area } A_2$$

$$= \frac{12}{5} \cdot 12 - \frac{8}{5} \cdot 8$$

$$= \frac{144 - 64}{10} = \frac{80}{10} = 8$$

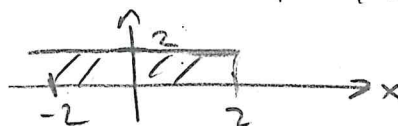


II. WAY

$$\int_{-2}^2 (2+5x) dx = \int_{-2}^2 2 dx + 5 \int_{-2}^2 x dx$$

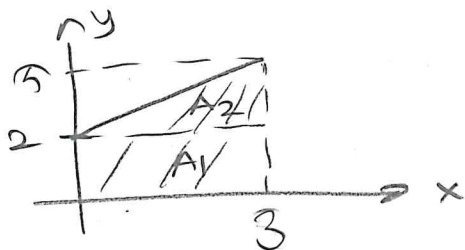
$$= 2 \cdot 4 + 0 = 8$$

$f(x)$ odd function
Integral is 0



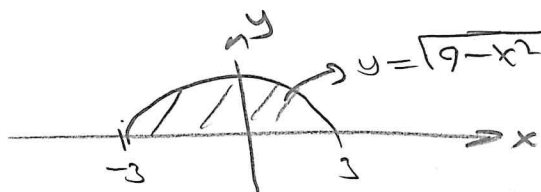
$$\textcircled{b} \int_0^3 (2+x) dx = \text{Area } A_1 + \text{Area } A_2$$

$$= 3 \cdot 2 + \frac{3 \cdot 3}{2} = \frac{21}{2}$$



$$\textcircled{c} \int_{-3}^3 \sqrt{9-x^2} dx = ?$$

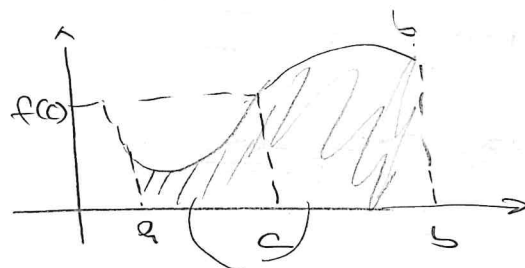
$$\int_{-3}^3 \sqrt{9-x^2} = 9\pi/2$$



THM A MEAN-VALUE THEOREM FOR INTEGRALS

If f is continuous on $[a, b]$, then there exists a point c in $[a, b]$ such that

$$\int_a^b f(x) dx = (b-a) f(c)$$



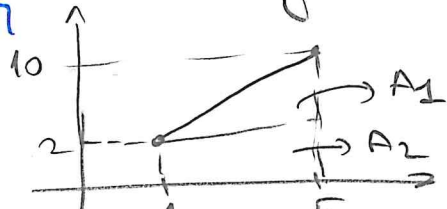
This area equals $= f(c) \cdot (b-a)$

DEFINITION Average value of a function

If f is integrable on $[a, b]$, then the average value or mean value of f on $[a, b]$ denoted by $\bar{f}$ is

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Ex Find average value of $f(x) = 2x$ on $[1, 5]$



$$\text{area } A_1 = 4 \cdot 2 = 8$$

$$+ \text{area } A_2 = 4 \cdot 8 = 16$$

(16)

$$\int_1^5 2x dx = 24$$

$$\bar{f} = \frac{1}{5-1} \cdot \int_1^5 2x dx = \frac{24}{4} = 6$$

Properties of Definite Integral

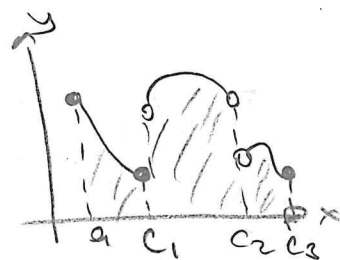
Piecewise Continuous Function

Let $c_0 < c_1 < c_2 < \dots < c_n$ be a finite set of points on the real line. A function f defined on $[c_0, c_n]$ except possibly at some points c_p , ($0 \leq p \leq n$), is called piecewise continuous on that interval if for each p ($1 \leq p \leq n$) there exists a function F_p continuous on the closed interval $[c_{p-1}, c_p]$ such that

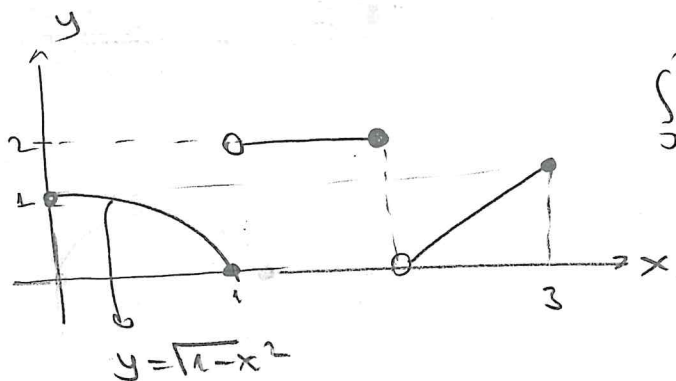
$$f(x) = F_p(x) \quad \text{on the open interval } (c_{p-1}, c_p)$$

In this case

$$\int_{c_0}^{c_n} f(x) dx = \sum_{p=1}^n \int_{c_{p-1}}^{c_p} F_p(x) dx$$



Ex Find $\int_0^3 f(x) dx$ where $f(x) = \begin{cases} \sqrt{1-x^2} & 0 \leq x \leq 1 \\ 2 & 1 < x \leq 2 \\ x-2 & 2 < x \leq 3 \end{cases}$



$$\begin{aligned} \int_0^3 f(x) dx &= \int_0^1 \sqrt{1-x^2} dx + \int_1^2 2 dx + \int_2^3 (x-2) dx \\ &= \frac{\pi}{4} + (2 \cdot 1) + \left(\frac{1}{2} \cdot 1 \cdot 1\right) \\ &= \frac{\pi+10}{4} \end{aligned}$$

THE FUNDAMENTAL THEOREM OF CALCULUS

THM The Fundamental Theorem of Calculus

Suppose that the function f is continuous on an interval I containing the point a .

PART I Let the function F be defined on I by

$$F(x) = \int_a^x f(t) dt$$

Then, F is differentiable on I , and $F'(x) = f(x)$ there. Thus, F is an antiderivative of f on I .

$$\frac{d}{dx} \underbrace{\int_a^x f(t) dt}_{F(x)} = f(x)$$

PART II If $G(x)$ is any antiderivative of $f(x)$ on I , so that $G'(x) = f(x)$ on I , then for any b in I we have

$$\int_a^b f(x) dx = G(b) - G(a)$$

Definition To facilitate the evaluation of definite integrals using the Fundamental Theorem of Calculus, we define the evaluation symbol

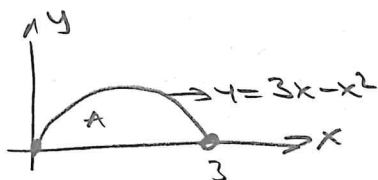
$$F(x) \Big|_a^b = F(b) - F(a)$$

Ex Evaluate (a) $\int_0^9 x^2 dx = ?$ (b) $\int_{-1}^2 (x^2 - 3x + 2) dx$

$$(a) \int_0^9 x^2 dx = \frac{x^3}{3} \Big|_0^9 = \frac{9^3}{3} - \frac{0^3}{3} = \frac{9^3}{3}$$

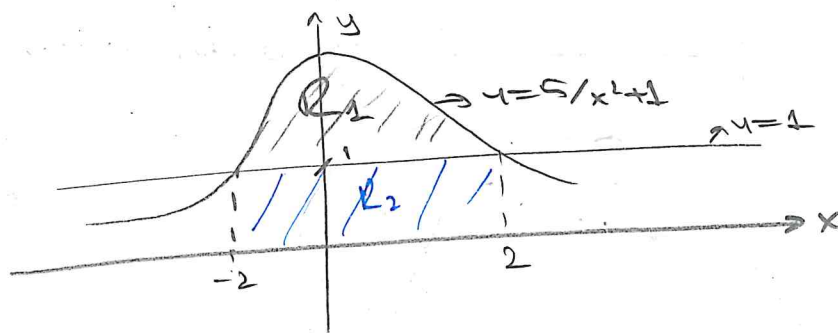
$$(b) \int_{-1}^2 (x^2 - 3x + 2) dx = \frac{x^3}{3} - 3\frac{x^2}{2} + 2x \Big|_{-1}^2 = \left(\frac{8}{3} - 6 + 4\right) - \left(-\frac{1}{3} - \frac{3}{2} - 2\right) = \frac{17}{6}$$

Ex 7 Find the area A of the plane region lying above the x-axis and under the curve $y = 3x - x^2$



$$A = \int_0^3 (3x - x^2) dx = \left[\frac{3}{2}x^2 - \frac{x^3}{3} \right]_0^3 = \frac{27}{2} - \frac{27}{3} - 0 = \frac{27}{6} = \frac{9}{2}$$

Ex 7 Find the area of the region lying above the line $y = 1$ and below the curve $y = \frac{5}{x^2 + 1}$



$$A = \int_{-2}^2 \frac{5}{x^2 + 1} dx - 4 = \left[5 \arctan x \right]_{-2}^2 - 4 = 10 \arctan 2 - 4$$

$$A = 2 \int_0^2 \frac{5}{x^2 + 1} dx - 4$$

even func.

$$A = 2 \int_0^2 \frac{5}{x^2 + 1} dx - 4 = 10 \arctan x \Big|_0^2 - 4 = 10 \arctan 2 - 4 \text{ square units.}$$

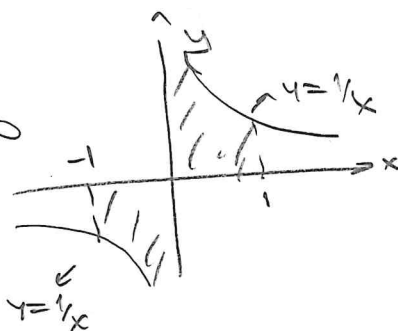
Ex 7 We know that $\frac{d}{dx} \ln|x| = \frac{1}{x}$ if $x \neq 0$. It is incorrect however, to state that

$$\int_{-1}^1 \frac{dx}{x} = \ln|x| \Big|_{-1}^1 = 0 - 0 = 0 \text{ even though } 1/x \text{ is an}$$

odd function. In fact, $1/x$ is undefined and has no limit at $x=0$ and not integrable on $[-1, 0]$ or $[0, 1]$. Observe that

$$\lim_{c \rightarrow 0^+} \int_c^1 \frac{1}{x} dx = \lim_{c \rightarrow 0^+} -\ln c = \infty$$

So both regions have infinite area
IMPROPER INTEGRALS



Ex Find the derivatives of the following functions.

(a) $F(x) = \int_x^3 e^{-t^2} dt$

(b) $G(x) = x^2 \int_{-4}^{5x} e^{-t^2} dt$

(c) $H(x) = \int_{x^2}^{x^3} e^{-t^2} dt$

(a) $F(x) = \int_x^3 e^{-t^2} dt = - \int_3^x e^{-t^2} dt \quad F'(x) = -e^{-x^2}$

(b) $G'(x) = 2x \int_{-4}^{5x} e^{-t^2} dt + x^2 \cdot \frac{d}{dx} \left(\int_{-4}^{5x} e^{-t^2} dt \right)$
 $= 2x \int_{-4}^{5x} e^{-t^2} dt + x^2 \cdot e^{-25x^2} (5)$
 $= 2x \int_{-4}^{5x} e^{-t^2} dt + 5x^2 e^{-25x^2}$

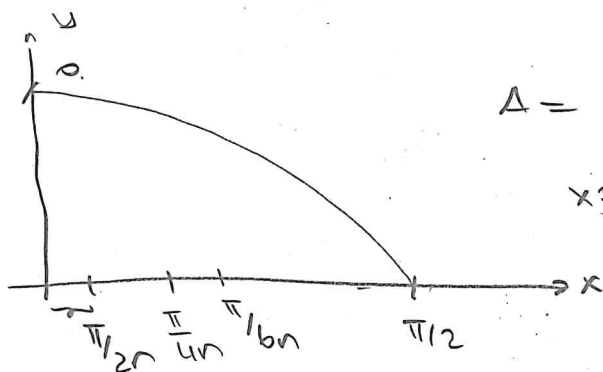
(c) $H(x) = \int_{x^2}^{x^3} e^{-t^2} dt = \int_{x^2}^0 e^{-t^2} dt + \int_0^{x^3} e^{-t^2} dt$
 $= - \int_0^{x^2} e^{-t^2} dt + \int_0^{x^3} e^{-t^2} dt$

$H'(x) = -e^{-(x^2)^2} \cdot 2x + e^{-(x^3)^2} \cdot 3x^2$

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) g'(x)$$

$$\frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt = f(g(x)) g'(x) - f(h(x)) h'(x)$$

Ex 7 Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \cos\left(\frac{j\pi}{2n}\right)$



$$\Delta = \frac{\pi/2 - 0}{n} = \frac{\pi}{2n}$$

$$x_j^* = 0 + j \frac{\pi}{2n} = j \cdot \frac{\pi}{2n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \cos\left(\frac{j\pi}{2n}\right) = \int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = 1$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \cos\left(\frac{j\pi}{2n}\right) &= \frac{2}{\pi} \lim_{n \rightarrow \infty} \frac{\pi}{2n} \sum_{j=1}^n \cos\left(\frac{j\pi}{2n}\right) \\ &= \frac{2}{\pi} \int_0^{\pi/2} \cos x dx = \frac{2}{\pi} \end{aligned}$$

THE METHOD OF SUBSTITUTION

Definition The indefinite integral of $f(x)$ on interval I is

$$\int f(x) dx = F(x) + C \text{ on } I \text{ provided that } F'(x) = f(x) \text{ for all } x \in I$$

Some Elementary Integrals

(1) $\int 1 dx = x + C$

(3) $\int x^2 dx = \frac{x^3}{3} + C$

(5) $\int \sqrt{x} dx = \frac{2}{3} x^{3/2} + C$

(7) $\int x^r dx = \frac{1}{r+1} x^{r+1} + C \quad r \neq -1$

(9) $\int \sin ax dx = -\frac{1}{a} \cos ax + C$

(11) $\int \sec^2(ax) dx = \frac{1}{a} \tan(ax) + C$

(13) $\int \sec(ax) \tan(ax) dx = \frac{1}{a} \sec(ax) + C$

(2) $\int x dx = \frac{x^2}{2} + C$

(4) $\int \frac{1}{x^2} dx = -\frac{1}{x} + C$

(6) $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$

(8) $\int \frac{1}{x} dx = \ln|x| + C$

(10) $\int \cos ax dx = \frac{1}{a} \sin ax + C$

(12) $\int \csc^2(ax) dx = -\frac{1}{a} \cot(ax) + C$

(14) $\int \csc(ax) \cot(ax) dx = -\frac{1}{a} \ln|\csc(ax) + \cot(ax)| + C$

$$(15) \int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C \quad (a > 0)$$

$$(16) \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$(17) \int e^{ax} dx = \frac{e^{ax}}{a} + C$$

$$(18) \int b^{ax} dx = \frac{1}{a \ln b} b^{ax}$$

$$(19) \int \cosh(ax) dx = \frac{1}{a} \sinh(ax) + C$$

$$(20) \int \sinh(ax) dx = \frac{1}{a} \cosh(ax) + C$$

$$\text{Ex } \int (x^4 - 3x^3 + 8x^2 - 6x - 7) dx = \frac{x^5}{5} - \frac{3x^4}{4} + \frac{8x^3}{3} - 3x^2 - 7x + C$$

$$\int \left(5x^{3/5} - \frac{3}{2+x^2} \right) dx = \frac{25}{8} x^{8/5} - \frac{3}{12} \arctan\left(\frac{x}{12}\right) + C$$

$$\int (4 \cos 5x - 5 \sin 3x) dx = \frac{4}{5} \sin 5x + \frac{5}{3} \cos 3x + C$$

$$\int \left(\frac{1}{\pi x} + a^{\pi x} \right) dx = \frac{1}{\pi} \ln|x| + \frac{1}{\pi \ln a} a^{\pi x} + C \quad (a > 0)$$

$$\begin{aligned} \text{Ex } \int \frac{(x+1)^2}{x} dx &= \int \frac{x^2 + 2x + 1}{x} dx \\ &= \int \left(x^2 + 2x + 1 + \frac{1}{x} \right) dx \\ &= \frac{x^3}{3} + \frac{2x^2}{2} + 3x + \ln|x| + C \end{aligned}$$

Method of Substitution

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x) \longrightarrow \int f'(g(x)) g'(x) dx = f(g(x)) + C$$

Let $u = g(x)$. Then, $\frac{du}{dx} = g'(x)$ or in differential form

$du = g'(x) dx$. Thus,

$$\int \underbrace{f'(g(x))}_u \underbrace{g'(x) dx}_{du} = \int f'(u) du = f(u) + C = f(g(x)) + C$$

Ex Find the following indefinite integrals.

① $\int \frac{x}{x^2+1} dx = ?$ $x^2+1 = u$
 $2x dx = du \Rightarrow x dx = \frac{du}{2}$

$$\int \frac{x}{x^2+1} dx = \int \frac{\frac{du}{2}}{u} = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|x^2+1| + C$$

② $\int \frac{\sin(3 \ln x)}{x} dx = ?$ $\ln x = u$
 $\frac{1}{x} dx = du$

$$\int \sin(3u) du = -\frac{\cos(3u)}{3} + C = -\frac{\cos(3 \ln x)}{3} + C$$

③ $\int e^x \sqrt{1+e^x} dx = \int v^{1/2} dv = \frac{v^{3/2}}{3/2} + C$

$$v = 1+e^x$$

$$e^x dx = dv$$

$$= \frac{2}{3} v^{3/2} + C$$

$$= \frac{2}{3} (1+e^x)^{3/2} + C$$

Ex Evaluate

$$\begin{aligned} \textcircled{1} \int \frac{dx}{x^2+4x+5} &= \int \frac{dx}{(x+2)^2+1} & t=x+2 \\ & & dt=dx \\ &= \int \frac{dt}{t^2+1} = \arctan t + C = \arctan(x+2) + C \end{aligned}$$

$$\begin{aligned} \textcircled{2} \int \frac{dx}{\sqrt{e^{2x}-1}} &= ? & \int \frac{dx}{\sqrt{e^{2x}-1}} &= \int \frac{dx}{\sqrt{e^{2x}(1-e^{-2x})}} = \int \frac{dx}{e^x \sqrt{1-(e^{-x})^2}} \\ u=e^{-x} & & & \\ du = -e^{-x} dx &= -\frac{dx}{e^x} & & \\ & & & \\ & & & = \int \frac{-du}{\sqrt{1-u^2}} = -\arcsin u + C \\ & & & = -\arcsin(e^{-x}) + C \end{aligned}$$

THM SUBSTITUTION IN A DEFINITE INTEGRAL

Suppose that g is a differentiable function on $[a, b]$ that satisfies $g(a)=A$ and $g(b)=B$. Also suppose that f is continuous on the range of g . Then,

$$\int_a^b f(g(x))g'(x)dx = \int_A^B f(u)du$$

Ex Evaluate the integral $I = \int_0^8 \frac{\cos(\sqrt{x+1})}{\sqrt{x+1}} dx$

METHOD I $u = \sqrt{x+1}$ $dx = \frac{dx}{2\sqrt{x+1}}$ if $x=0, u=1$
 $x=8, u=3$

$$I = 2 \int_1^3 \cos u du = 2 \sin u \Big|_1^3 = 2 \sin 3 - 2 \sin 1$$

METHOD II $I = 2 \int_{x=0}^{x=8} \cos u du = 2 \sin u \Big|_{x=0}^{x=8} = 2 \sin(\sqrt{x+1}) \Big|_0^8$
 $= 2 \sin 3 - 2 \sin 1$

Ex 7 Find the area of the region bounded by

$$y = \left(2 + \sin\left(\frac{x}{2}\right)\right)^2 \cos\left(\frac{x}{2}\right), \text{ the } x \text{ axis, and the lines } x=0 \text{ and } x=\pi$$

Since $y > 0$ when $0 \leq x \leq \pi$, the required area is

$$A = \int_0^{\pi} \left(2 + \sin\left(\frac{x}{2}\right)\right)^2 \cos\left(\frac{x}{2}\right) dx$$

$$v = 2 + \sin\left(\frac{x}{2}\right)$$

$$dv = \frac{1}{2} \cos\left(\frac{x}{2}\right) dx$$

$$= 2 \int_2^3 v^2 dv = \frac{2}{3} v^3 \Big|_2^3 = \frac{2}{3} (27 - 8) = \frac{38}{3}$$

TRIGONOMETRIC INTEGRALS

① $\int \tan x dx = \ln |\sec x| + C$

② $\int \cot x dx = \ln |\sin x| + C = -\ln |\csc x| + C$

③ $\int \sec x dx = \ln |\sec x + \tan x| + C$

④ $\int \csc x dx = -\ln |\csc x + \cot x| + C$
 $= \ln |\csc x - \cot x| + C$

Proof ① $\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \int -\frac{du}{u} = -\ln |u| + C$
 $\cos x = u \quad -\sin x dx = du$

$$\int \tan x dx = -\ln |\cos x| + C = \ln \left| \frac{1}{\cos x} \right| + C = \ln |\sec x| + C$$

② $\int \cot x dx = \int \frac{\cos x}{\sin x} dx = \int \frac{du}{u} = \ln |u| + C = \ln |\sin x| + C$
 $\sin x = u \quad \cos x dx = du$

③ $\int \sec x dx = \int \sec x \cdot \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx$ $\sec x + \tan x = u$
 $(\sec^2 x + \sec x \cdot \tan x) dx = du$

$$= \int \frac{du}{u} = \ln |\sec x + \tan x| + C$$

④ $\int \csc x dx = \int \csc x \cdot \frac{(\csc x + \cot x)}{(\csc x + \cot x)} dx$ $(\csc x + \cot x)' = -\csc x \cot x - \csc^2 x$

$$= -\ln |\csc x + \cot x| + C$$

$\int \sin^m x \cos^n x dx$; use method of substitution, if m or n odd

Ex 7/7 (a) $\int \sin^3 x \cos^8 x dx = ?$

$$\int \sin x \cdot \sin^2 x \cos^8 x dx = \int u^8 (1-u^2) (-du)$$

$$\cos x = u \quad \sin^2 x = 1 - \cos^2 x$$

$$-\sin x dx = du$$

$$= -\int (u^8 - u^{10}) du = -\frac{u^9}{9} + \frac{u^{11}}{11} + C$$

$$= -\frac{(\cos x)^9}{9} + \frac{(\cos x)^{11}}{11} + C$$

(b) $\int \cos^5 ax dx = ?$

$$\int \cos(ax) \cos^4(ax) dx \quad \frac{du}{a} = \int \frac{du}{a} (1-u^2)^2$$

$$\sin(ax) = u \Rightarrow \cos^2(ax) = 1 - u^2$$

$$\cos(ax) \cdot a dx = du$$

$$= \int \frac{du}{a} (1 - 2u^2 + u^4) = \frac{1}{a} \left(u - 2\frac{u^3}{3} + \frac{u^5}{5} \right) + C$$

$$= \frac{1}{a} \left(\sin(ax) - \frac{2}{3}(\sin(ax))^3 + \frac{(\sin(ax))^5}{5} \right) + C$$

$\int \sin^m x \cos^n x dx$ powers of m and n are both odd use double angle formulas

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x) \quad \text{and} \quad \sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

Ex 7/7 $\int \sin^4 x dx = ?$

$$\int \sin^4 x dx = \frac{1}{4} \int (1 - \cos 2x)^2 dx$$

$$= \frac{1}{4} \int (1 - 2\cos 2x + (\cos 2x)^2) dx$$

$$= \frac{x}{4} - \frac{1}{4} \sin 2x + \frac{1}{8} \int (1 + \cos 4x) dx$$

$$= \frac{x}{4} - \frac{1}{4} \sin 2x + \frac{x}{8} + \frac{1}{32} \sin 4x + C$$

$$\int \sec^n x \tan^n x dx \quad \text{or} \quad \int \csc^n x \cot^n x dx \quad \text{unless mis add and n is even}$$

Use $\sec^2 x = 1 + \tan^2 x$ $\csc^2 x = 1 + \cot^2 x$

Ex ① $\int \tan^2 x dx = ?$ $\int \tan^2 x dx = \int (\sec^2 x - 1) dx = \tan x - x + C$

$$1 + \tan^2 x = \sec^2 x$$

② $\int \sec^4 t dt = \int (1 + \tan^2 t) \underbrace{\sec^2 t dt}_{du} = \int (1 + u^2) du \quad \begin{matrix} u = \tan t \\ du = \sec^2 t dt \end{matrix}$

$$= u + \frac{u^3}{3} + C = \tan t + \frac{(\tan t)^3}{3} + C$$

③ $\int \sec^3 x \tan^3 x dx = \int \sec^2 x \underbrace{(\sec^2 x - 1)}_{\tan^2 x} \overbrace{\sec x \tan x dx}^{du}$ $\begin{matrix} u = \sec x \\ du = \sec x \tan x dx \end{matrix}$

$$= \int (u^4 - u^2) du = \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{(\sec x)^5}{5} - \frac{(\sec x)^3}{3} + C$$

EXAMPLES

RIEMMAN SUMS

Ex Obtain the area under the given functions using Riemann Sums.

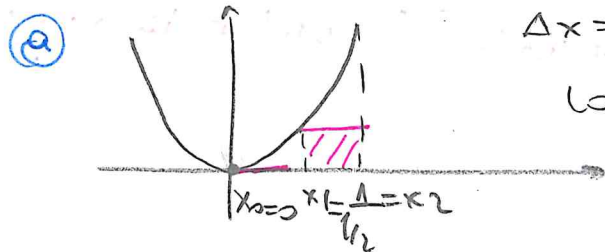
- Equally length, lower Riemann sum $n=2$.
- Equally length, lower Riemann sum $n=4$
- Equally length, upper Riemann sum $n=2$
- Equally length, upper Riemann sum $n=4$
- Middle point $n=2$.

① $f(x) = x^2 \quad x \in [0, 1]$

② $f(x) = 4 - x^2 \quad x \in [-2, 2]$

③ $f(x) = x^3 \quad x \in [0, 1]$

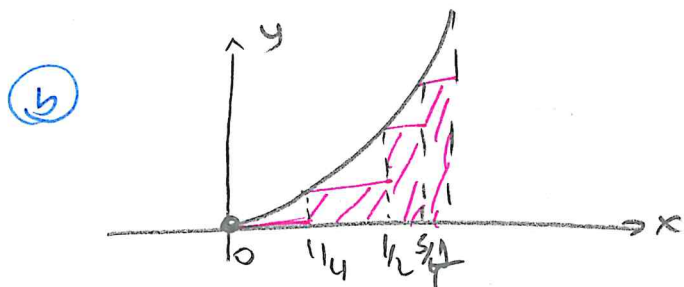
① $f(x) = x^2$ $x \in [0, 1]$



$\Delta x = \frac{1-0}{2}$ $x_i = x_0 + i \Delta x = i/2$ $x_0=0$ $x_1=1/2$ $x_2=1$

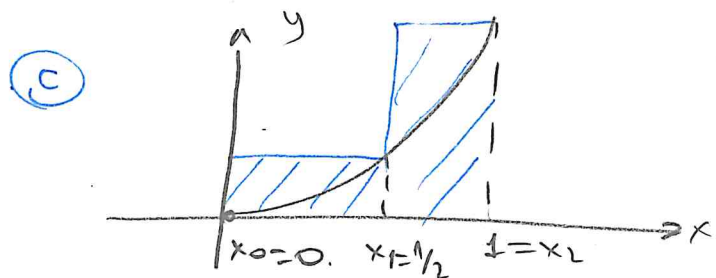
lower sum $\Rightarrow$ evaluate at x_{i-1}

$L = \sum_{i=0}^1 \left(\frac{i}{2}\right)^2 \cdot \frac{1}{2} = \frac{1}{2} \left(0 + \left(\frac{1}{2}\right)^2\right) = \frac{1}{8}$



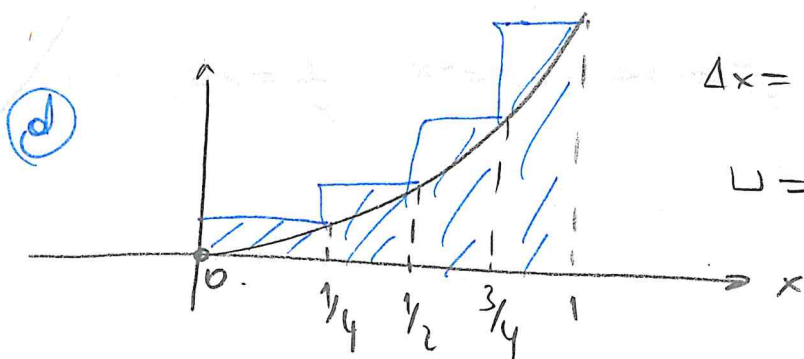
$\Delta x = \frac{1-0}{4}$ $x_i = x_0 + i \Delta x = i/4$

$L = \sum_{i=0}^3 \left(\frac{i}{4}\right)^2 \cdot \frac{1}{4}$
 $= \frac{1}{4} \left(0 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2\right) = \frac{7}{32}$



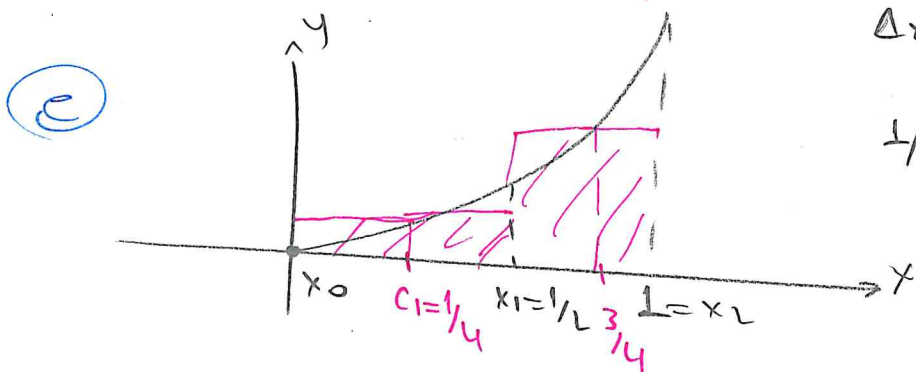
$\Delta x = \frac{1}{2}$ $x_i = x_0 + i \Delta x = i/2$

$L = \sum_{i=0}^1 \left(\frac{i}{2}\right)^2 \cdot \frac{1}{2} = \frac{1}{2} \left(\left(\frac{1}{2}\right)^2 + 1^2\right) = \frac{5}{8}$



$\Delta x = \frac{1}{4}$ $x_i = x_0 + i \Delta x = i/4$

$L = \sum_{i=0}^3 \left(\frac{i}{4}\right)^2 \cdot \frac{1}{4} = \frac{1}{4} \left[\left(\frac{1}{4}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2 + 1^2\right]$

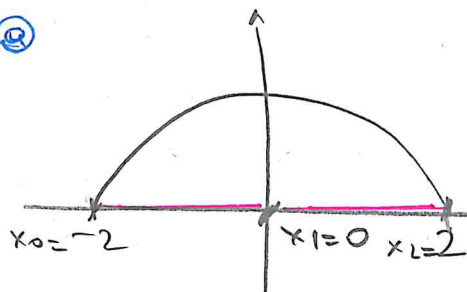


$\Delta x = \frac{1}{2}$

$\frac{1}{2} (f(1/4) + f(3/4)) = \frac{5}{16}$

2) $f(x) = 4 - x^2 \quad x \in [-2, 2]$

a)

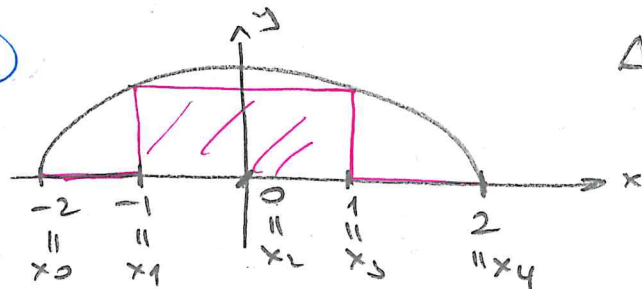


$$\Delta x = \frac{2 - (-2)}{2} = 2.$$

$$x_p = x_0 + p \cdot \Delta x = -2 + 2p$$

$$L = 2((4 - (-2)^2) + 2(4 - 2^2)) = 0$$

b)

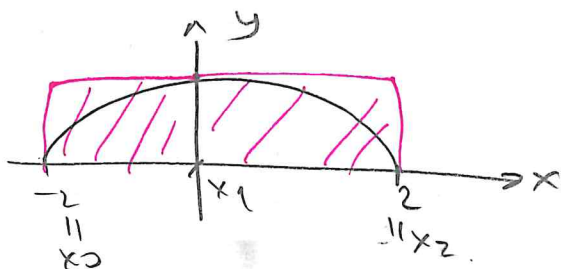


$$\Delta x = \frac{2 - (-2)}{4} = 1 \quad x_p = -2 + p \Delta x = -2 + p$$

$$\frac{1}{2} \sum_{p=0}^4 (4 - x_p^2) \cdot 1 + \sum_{p=3}^4 (4 - x_p^2) \cdot 1$$

$$= (4 - (-2)^2) + (4 - (-1)^2) = 6.$$

c)

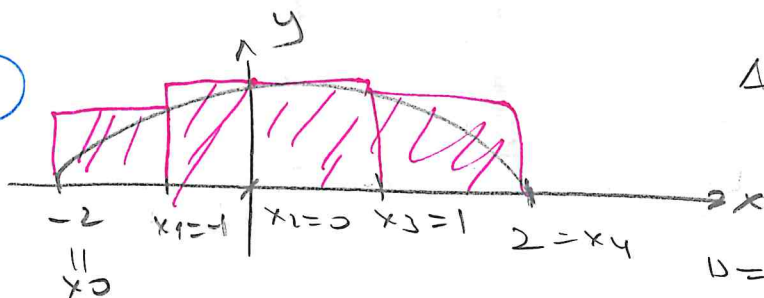


$$\Delta x = \frac{2 - (-2)}{2} = 2$$

$$x_p = x_0 + p \Delta x = -2 + 2p$$

$$U = 2(4 - 0^2) + 2(4 - 0^2) = 16$$

d)



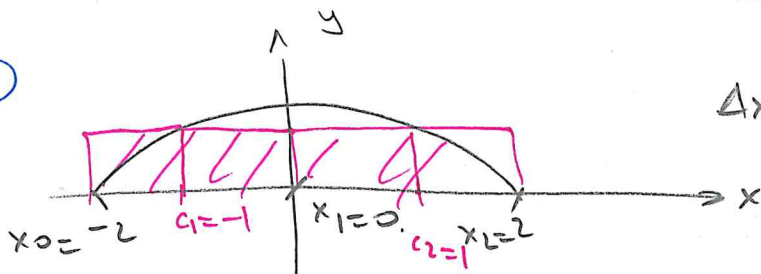
$$\Delta x = \frac{2 - (-2)}{4} = 1$$

$$x_p = -2 + p \Delta x = -2 + p$$

$$U = \sum_{p=1}^3 (-4 - x_p^2) \cdot 1 + \sum_{p=2}^3 (4 - x_p^2) \cdot 1$$

$$= (-4 - (-1)^2) + (4 - 0^2) + (4 - 0^2) + (4 - 1^2) =$$

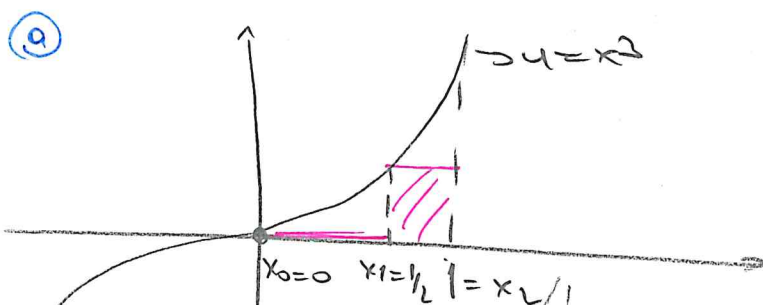
e)



$$\Delta x = 2 \quad M = 2[f(-1) + f(1)]$$

$$= 2[3 + 3] = 12$$

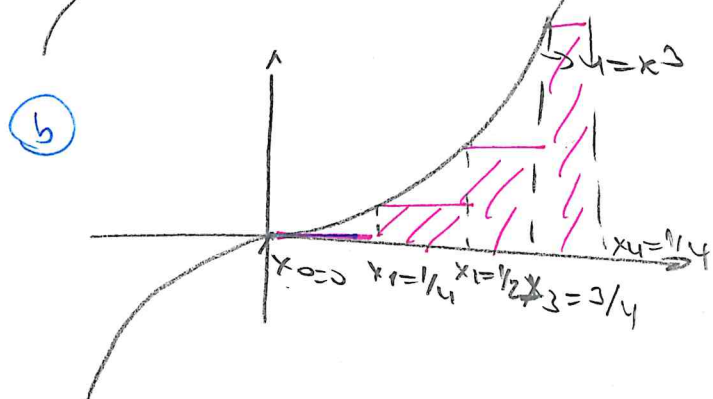
③ $f(x) = x^3 \quad x \in [0, 1]$



$$\Delta x = \frac{1-0}{2} = \frac{1}{2}$$

$$x_p = 0 + p/2 = p/2$$

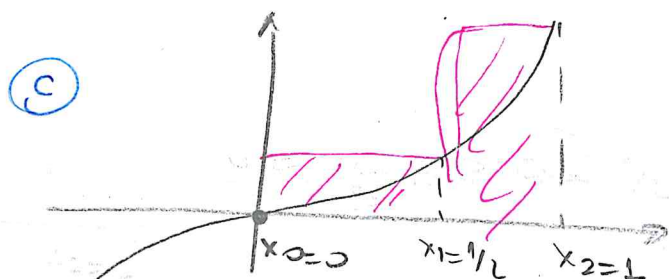
$$L = \sum_{p=0}^1 f(x_p) \Delta x = \frac{1}{2} (0 + (1/2)^3) = \frac{1}{16}$$



$$\Delta x = \frac{1-0}{4} = 1/4$$

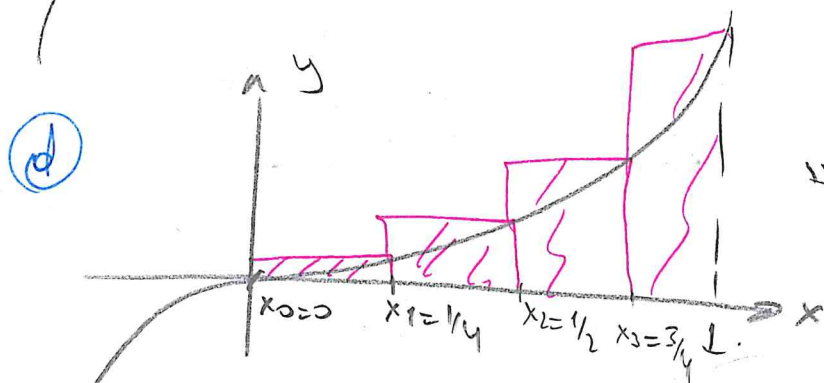
$$x_p = 0 + p/4$$

$$L = \sum_{p=0}^3 f(x_p) \Delta x = \frac{1}{4} (0 + (1/4)^3 + (1/2)^3 + (3/4)^3) = 9/64$$



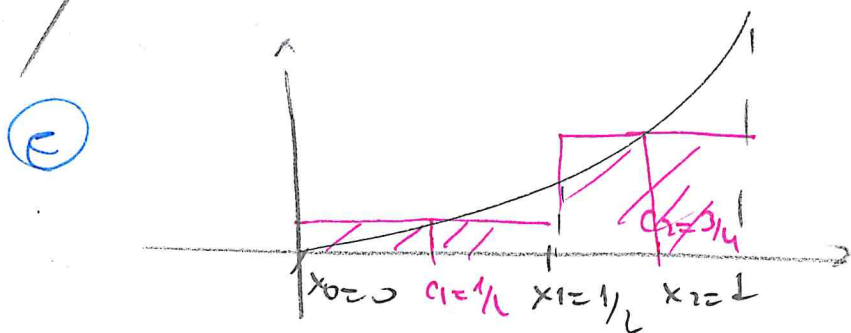
$$\Delta x = \frac{1-0}{2} = 1/2 \quad x_p = 0 + p/2$$

$$L = \sum_{p=0}^1 f(x_p) \Delta x = \frac{1}{2} ((1/2)^3 + 1^3) = \frac{9}{16}$$



$$\Delta x = \frac{1-0}{4} \quad x_p = 0 + p/4$$

$$L = \sum_{p=0}^4 f(x_p) \Delta x = \frac{1}{4} ((1/4)^3 + (1/2)^3 + (3/4)^3 + 1^3) = \frac{25}{64}$$



$$\Delta x = 1/2$$

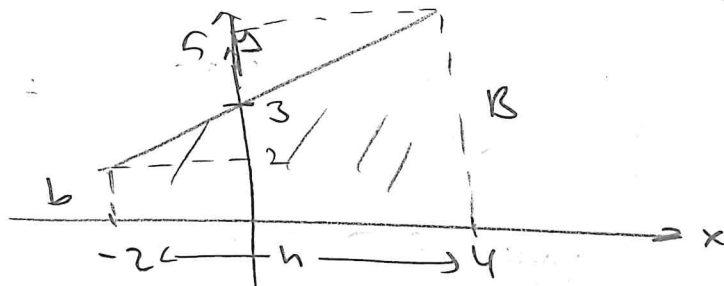
$$\frac{1}{2} [f(1/4) + f(1/2)]$$

$$= \frac{1}{2} ((1/4)^3 + (1/2)^3)$$

$$= 7/32$$

Ex Use areas of regions to evaluate the following integrals:

(a) $\int_{-2}^4 \left(\frac{x}{2} + 3\right) dx = ?$

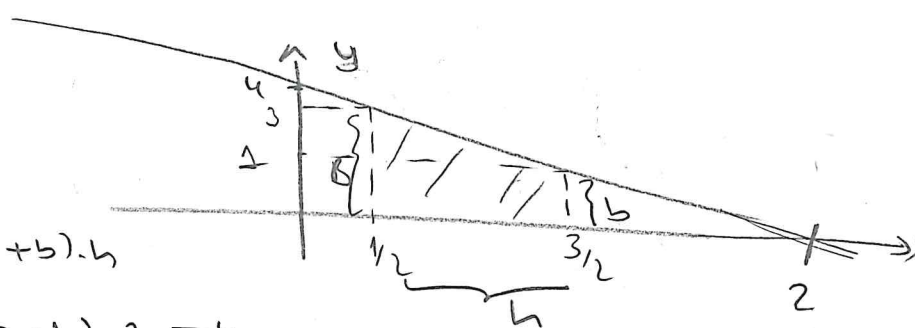


$$\text{Area} = \frac{1}{2} (B+b) \cdot h = \frac{1}{2} (5+2) \cdot 6 = 21 = \int_{-2}^4 \left(\frac{x}{2} + 3\right) dx$$

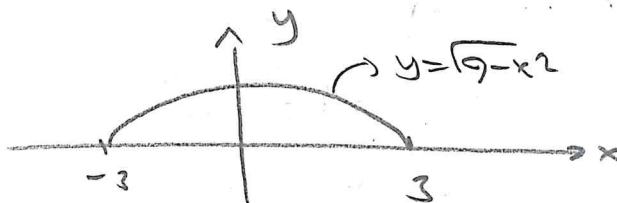
(b) $\int_{1/2}^{3/2} (-2x+4) dx = ?$

$$\int_{1/2}^{3/2} (-2x+4) dx = \frac{1}{2} (B+b) \cdot h$$

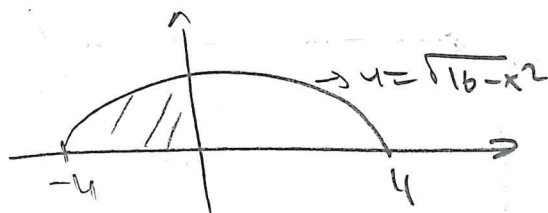
$$= \frac{1}{2} (3+1) \cdot 2 = 4$$



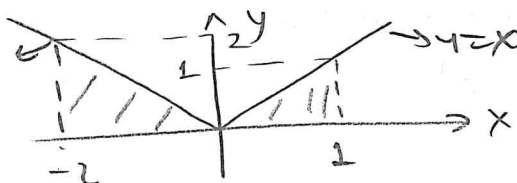
(c) $\int_{-3}^3 \sqrt{9-x^2} dx = \frac{1}{2} \pi r^2$
 $= \frac{1}{2} \pi \cdot 9$



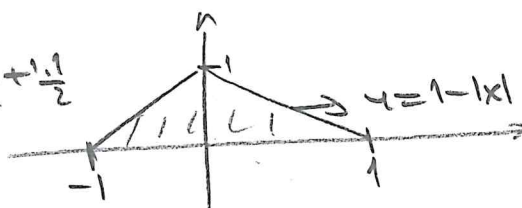
(d) $\int_{-4}^0 \sqrt{16-x^2} dx = \frac{\pi \cdot 16}{4} = 4\pi$



(e) $\int_{-2}^1 |x| dx = \frac{2 \cdot 2}{2} + \frac{1 \cdot 1}{2} = \frac{5}{2}$



(f) $\int_{-1}^1 (1-|x|) dx = 1 = \frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}$



Ex 7 Represent the following summations using integrals.

(a) $I = \lim_{\substack{\|P\| \rightarrow 0 \\ n \rightarrow \infty}} \sum_{k=1}^n c_k^2 \Delta x_k$ P is the partition of $[0, 2]$

$$I = \int_0^2 x^2 dx.$$

(b) $I = \lim_{\substack{\|P\| \rightarrow 0 \\ n \rightarrow \infty}} \sum_{k=1}^n \frac{1}{1-c_k} \Delta x_k$ P is the partition of $[2, 3]$

$$I = \int_2^3 \frac{1}{1-x} dx$$

(c) $I = \lim_{\substack{\|P\| \rightarrow 0 \\ n \rightarrow \infty}} \sum_{k=1}^n \tan(c_k) \Delta x_k$ P is the partition of $[0, \pi/4]$

$$I = \int_0^{\pi/4} \tan x dx$$

Ex 8 Evaluate the following integrals by using the definition.

$$I = \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k$$

NOTE Here, since $\|P\| \rightarrow 0$ c_k 's are not important. Therefore, we will use equally spaced points i.e. $\Delta x_k = \Delta x$ $k=1, \dots, n$

(a) $\int_0^b x dx = ?$ $b > 0$

$$\Delta x = \frac{b-0}{n} = \frac{b}{n} \quad P = \left\{ 0, \frac{b}{n}, \frac{2b}{n}, \dots, \frac{nb}{n} \right\}$$

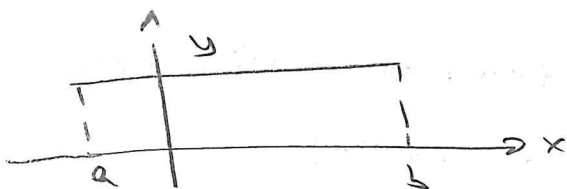
$$c_k = \frac{k \cdot b}{n} \quad f(c_k) = \frac{k \cdot b}{n}$$

$$\sum_{k=1}^n f(c_k) \Delta x = \sum_{k=1}^n \frac{k \cdot b}{n} \cdot \frac{b}{n} = \frac{b^2}{n^2} \sum_{k=1}^n k = \frac{b^2}{n^2} \cdot n \cdot \frac{(n+1)}{2}$$

$$\lim_{n \rightarrow \infty} \frac{b^2}{n^2} \cdot \frac{n(n+1)}{2} = \frac{b^2}{2}$$

$$(b) I = \int_a^b c \, dx = ?$$

$$a < 0 \\ b > 0$$



$$\Delta x = \frac{b-a}{n} \quad c_k = a + k \left(\frac{b-a}{n} \right)$$

$$\sum_{k=1}^n \Delta x \cdot f(c_k) = \sum_{k=1}^n \frac{b-a}{n} \cdot c = c \cdot \frac{(b-a)}{n} \cdot \sum_{k=1}^n 1 = c(b-a)$$

$$\lim_{n \rightarrow \infty} c(b-a) = c(b-a)$$

$$(c) I = \int_0^2 (2x+1) \, dx$$

$$\Delta x = \frac{2}{n} \quad c_k = 0 + \frac{2}{n} \cdot k \quad f(c_k) = \frac{4k}{n} + 1$$

$$I = \sum_{k=1}^n \left(\frac{4k}{n} + 1 \right) \cdot \frac{2}{n} = \frac{8}{n^2} \sum_{k=1}^n k + \frac{2}{n} \sum_{k=1}^n 1$$

$$= \frac{8}{n^2} \cdot \frac{n(n+1)}{2} + \frac{2}{n} \cdot n$$

$$= 4 \frac{n^2+n}{n^2} + 2$$

$$\lim_{n \rightarrow \infty} 4 \left(\frac{n^2+n}{n^2} \right) + 2 = \lim_{n \rightarrow \infty} 4 \left(1 + \frac{1}{n} \right) + 2 = 6$$

$$(d) \int_0^b x^2 \, dx = ?$$

$$\Delta x = \frac{b-a}{n} \quad c_k = a + \left(\frac{b-a}{n} \right) k$$

$$f(c_k) = \left(a + \left(\frac{b-a}{n} \right) k \right)^2$$

$$S_n = \sum_{k=1}^n f(c_k) \Delta x = \sum_{k=1}^n \left(a + \left(\frac{b-a}{n} \right) k \right)^2 \left(\frac{b-a}{n} \right)$$

$$= \frac{b-a}{n} \left[\sum_{k=1}^n a^2 + 2a \sum_{k=1}^n \left(\frac{b-a}{n} \right) k + \left(\frac{b-a}{n} \right)^2 \cdot \sum_{k=1}^n k^2 \right]$$

$$= \frac{b-a}{n} \left[a^2 n + 2a \left(\frac{b-a}{n} \right) \cdot \frac{n(n+1)}{2} + \left(\frac{b-a}{n} \right)^2 \cdot \frac{n \cdot (n+1)(2n+1)}{6} \right]$$

$$\lim_{n \rightarrow \infty} S_n = \frac{b^3}{3} - \frac{a^3}{3}$$

(31)

FUNDAMENTAL THEOREM OF CALCULUS

Ex Evaluate the following integrals?

$$(a) y = \int_0^x (t^3 + 1) dt \Rightarrow \frac{dy}{dx} = x^3 + 1$$

$$(b) y = \int_x^5 3t \sin t dt \Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\int_x^5 3t \sin t dt \right) \\ = \frac{d}{dx} \left(- \int_5^x 3t \sin t dt \right) \\ = -3x \sin x$$

$$(c) y = \int_1^{x^2} \cos t dt \Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\int_1^{x^2} \cos t dt \right) = \cos x^2 \cdot 2x$$

$$(d) y = \int_1^{\sin x} 3t^2 dt \Rightarrow \frac{dy}{dx} = 3(\sin x)^2 \cdot \cos x$$

$$(e) y = \int_{\sqrt{x}}^0 \sin t^2 dt \Rightarrow \frac{dy}{dx} = -\sin(\sqrt{x})^2 \cdot \frac{1}{2\sqrt{x}}$$

INTEGRALS

Ex Evaluate the following integrals.

$$(1) I = \int_1^4 \left(\frac{3}{2} \sqrt{x} - \frac{4}{x^2} \right) dx = ? \quad I = \frac{x^{3/2} \cdot \frac{3}{2}}{3/2} - \frac{4 \cdot x^{-2+1}}{-1} \Big|_1^4 \\ = [8+1] - 5 = 4$$

$$(2) \int_1^5 \frac{(x^2)^{1/3} - x}{x^5} dx = \int_1^5 (x^{-7/3} - x^{-2}) dx = \frac{x^{-7/3+1}}{-7/3+1} - \frac{x^{-2+1}}{-2+1} \Big|_1^5 \\ = \left(-\frac{3}{4} x^{-4/3} + \frac{1}{x} \right) \Big|_1^5 = -\frac{3}{4} \frac{1}{(625)^{1/3}} + \frac{1}{5} - \frac{1}{4}$$

$$(3) \int_0^1 (x^2+1)^2 dx = \int_0^1 (x^4 + 2x^2 + 1) dx = \frac{x^5}{5} + 2\frac{x^3}{3} + x \Big|_0^1 = \frac{1}{5} + \frac{2}{3} + 1 = \frac{18}{15} + \frac{20}{15} + \frac{15}{15} = \frac{53}{15}$$

$$(4) \int_{-1}^2 |x| dx = \int_{-1}^0 |x| dx + \int_0^2 |x| dx = \int_{-1}^0 (-x) dx + \int_0^2 x dx$$

$$= -\frac{x^2}{2} \Big|_{-1}^0 + \frac{x^2}{2} \Big|_0^2 = \frac{1}{2} + \frac{4}{2} = 2.5$$

$$(5) \int \sin^2 x dx = \int \frac{1 - \cos 2x}{2} dx = \frac{1}{2}x - \frac{\sin 2x}{4} + C$$

$$(6) \int \sin^2 3x dx = \int \frac{1 - \cos 6x}{2} dx = \frac{x}{2} - \frac{\sin 6x}{12} + C$$

$$(7) \int \sin x \cos x dx = \int \frac{\sin 2x}{2} dx = \frac{\cos 2x}{4} + C$$

$$(8) \int_0^1 \sinh^2 x dx = \int_0^1 \frac{\cosh 2x - 1}{2} dx$$

$$= \frac{1}{2} \int_0^1 (\cosh 2x - 1) dx = \frac{1}{2} \left[\frac{\sinh 2x}{2} - x \right]_0^1$$

$$= \frac{1}{4} \sinh 2 - \frac{1}{2} \approx 0.4067$$

$$(9) \int 2^{3x} \cdot 5^x dx = \int 8^x \cdot 5^x dx = \int 40^x dx$$

$$= \frac{40^x}{\ln 40} + C$$

$$(10) \int \frac{e^{-3x} - 2e^{2x}}{e^{5x}} dx = \int (e^{-8x} - 2e^{-3x}) dx$$

$$= -\frac{1}{8} e^{-8x} + \frac{2}{3} e^{-3x} + C$$

$$(11) \int_0^{\ln 2} 4 \cdot e^x \sinh x dx = \int_0^{\ln 2} 4 \cdot e^x \left(\frac{e^x - e^{-x}}{2} \right) dx$$

$$= \int_0^{\ln 2} 2(e^{2x} - 1) dx = (e^{2x} - 2x) \Big|_0^{\ln 2} = 4 - 2 \ln 2 - 1 \approx 1.613$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - \sin^2 x - \sin^2 x \\ &= 1 - 2\sin^2 x \end{aligned}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - (1 - \cos^2 x) \\ &= 2\cos^2 x - 1 \end{aligned}$$

$$\cos^2 x = \frac{\cos 2x + 1}{2}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\begin{aligned} \cosh 2x &= \cosh^2 x + \sinh^2 x \\ &= \cosh^2 x + \cosh^2 x - 1 \\ &= 2\cosh^2 x - 1 \end{aligned}$$

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}$$

$$\cosh 2x = 1 + \sinh^2 x + \sinh^2 x$$

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

$$(12) \int_0^{\pi} \cos x \, dx = \sin x \Big|_0^{\pi} = \sin \pi - \sin 0 = 0$$

$$(13) \int_{-\pi/4}^0 \sec x \tan x \, dx = \sec x \Big|_{-\pi/4}^0 = \sec(0) - \sec(-\pi/4) = 1 - \frac{1}{\sqrt{2}}$$

$$(14) \int_{\sqrt{2}/2}^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}} = \arcsin x \Big|_{\sqrt{2}/2}^{\sqrt{3}/2} = \arcsin\left(\frac{\sqrt{3}}{2}\right) - \arcsin\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

$$(15) \int_{\sqrt{3}/2}^{\sqrt{2}} \frac{dx}{x\sqrt{x^2-1}} = \operatorname{arcsec} x \Big|_{\sqrt{3}/2}^{\sqrt{2}} = \operatorname{arcsec}(\sqrt{2}) - \operatorname{arcsec}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

$$(16) \int \frac{dx}{\sqrt{9-x^2}} = \int \frac{dx}{\sqrt{3^2-x^2}} = \arcsin\left(\frac{x}{3}\right) + C$$

$$(17) \int \frac{dx}{\sqrt{3-4x^2}} = \int \frac{dx}{\sqrt{4\left(\frac{3}{4}-x^2\right)}} = \int \frac{dx}{2\sqrt{\frac{3}{4}-x^2}} = \frac{1}{2} \arcsin\left(\frac{x}{\sqrt{3}/2}\right) + C$$